

PRACTICAL 1

Output :

```
manasvi@ManasviG:~$ nano prac1.c
manasvi@ManasviG:~$ gcc prac1.c
manasvi@ManasviG:~$ ./a.out
Enter a Linux command: ls

--- Parent Process ---
Parent ID: 529
Parent's Parent ID: 328

--- Child Process ---
Child ID: 530
Parent ID: 529
a.out a8 b2 execl.c f1 f2.c f6 linux_task2 mvex.c open1.c pipe2.c pipeex3.c task1.c z7
a1 a9 dirz execlp.c f1.c f3 fork4.c linux_task3 mvex1 permission_lab pipeex1.c prac1.c task1.c.save z8
a7 b1 ex1 execl.c f2 f4 linux_task1 linux_task4 open.c pipe1.c pipeex2.c readwrite.c xyz
Child process has finished execution
manasvi@ManasviG:~$ |
```

b) Linux terminal commands

- uname

```
manasvi@ManasviG:~$ uname
Linux
manasvi@ManasviG:~$ |
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- lscpu

```
manasvi@ManasviG:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                       12
On-line CPU(s) list:         0-11
Vendor ID:                   GenuineIntel
Model name:                   13th Gen Intel(R) Core(TM) i5-1334U
CPU family:                   6
Model:                       186
Thread(s) per core:           2
Core(s) per socket:           6
Socket(s):                    1
Stepping:                     3
BogoMIPS:                     4992.02
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht
                             syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_kno
                             wn_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes x
                             save avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow
                             ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha
                             _ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir6
                             4b fsrm md_clear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:               VT-x
Hypervisor vendor:            Microsoft
Virtualization type:          full
Caches (sum of all):
L1d:                           288 KiB (6 instances)
L1i:                           192 KiB (6 instances)
L2:                            7.5 MiB (6 instances)
L3:                            12 MiB (1 instance)
NUMA:
NUMA node(s):                  1
```

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Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected
manasvi@ManasviG:~$ |

```

- lsblk

```

manasvi@ManasviG:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0     2G  0 disk [SWAP]
sdd   8:48   0     1T  0 disk /mnt/wslg/distro
/
manasvi@ManasviG:~$ |

```

- top

```

manasvi@ManasviG:~$ top
top - 16:56:12 up 12 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks: 24 total,  1 running, 23 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7784.6 total, 7201.5 free,  554.8 used, 179.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7229.9 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24556  15408 11484  S   0.0   0.2   0:00.51 systemd
    2 root        20   0    3180   2204  2072  S   0.0   0.0   0:00.00 init-systemd(Ub
    7 root        20   0    3180   2048  1940  S   0.0   0.0   0:00.00 init
   47 root        19  -1  50400  16636 15360  S   0.0   0.2   0:00.13 systemd-journal
   79 systemd+    20   0   22416  14484 11988  S   0.0   0.2   0:00.06 systemd-resolve
   86 root        20   0   35372  12328  9236  S   0.0   0.2   0:00.34 systemd-udev
  105 root        20   0    2888   1940  1816  S   0.0   0.0   0:00.03 chronyd-starter
  109 root        20   0    4472   3152  2908  S   0.0   0.0   0:00.00 cron
  114 message+   20   0    8820   5448  4772  S   0.0   0.1   0:00.03 dbus-daemon
  153 root        20   0   44096  29788 14648  S   0.0   0.4   0:00.19 networkd-dispat
  158 root        20   0   18436   9392  8148  S   0.0   0.1   0:00.07 systemd-logind
  182 syslog     20   0  220548   5552  4352  S   0.0   0.1   0:00.06 rsyslogd
  212 root        20   0    5492   2712  2528  S   0.0   0.0   0:00.00 agetty
  231 _chrony     20   0   23600  11220  7236  S   0.0   0.1   0:00.05 chronyd
  234 _chrony     20   0   12096   2468  1820  S   0.0   0.0   0:00.00 chronyd
  272 root        20   0  123460  32656 18040  S   0.0   0.4   0:00.11 unattended-upgr
  326 root        20   0    3184   1108    980  S   0.0   0.0   0:00.00 SessionLeader
  327 root        20   0    3200   1248   1108  S   0.0   0.0   0:00.04 Relay(328)
  328 manasvi    20   0    6268   5544  3864  S   0.0   0.1   0:00.04 bash
  329 root        20   0    8644   5628  4788  S   0.0   0.1   0:00.00 login
  372 manasvi    20   0   22344  13528 10900  S   0.0   0.2   0:00.08 systemd
  376 manasvi    20   0   22912   4136   2136  S   0.0   0.1   0:00.00 (sd-pam)
  402 manasvi    20   0    6136   5500   3928  S   0.0   0.1   0:00.02 bash
  678 manasvi    20   0    8164    6036  3940  R   0.0   0.1   0:00.00 top

```